Industrial Internship Report on

” Prediction of Agriculture Crop Production in India”

Prepared by

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Executive Summary
This report describes the work carried out during the 6-week industrial internship provided by Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd. The project focuses on predicting agricultural crop yield using machine learning techniques. Agriculture plays a vital role in the Indian economy, and improving crop productivity is essential for sustainable farming. This project utilizes agricultural data such as crop type, state, cultivation cost, production cost, and yield information to analyze patterns and predict crop yield accurately. Machine learning algorithms were applied to process and analyze the dataset. Data preprocessing, exploratory data analysis, feature encoding, and model training were performed to develop a predictive system. The developed model helps farmers and agricultural stakeholders make data-driven decisions, optimize resource utilization, and improve agricultural productivity.

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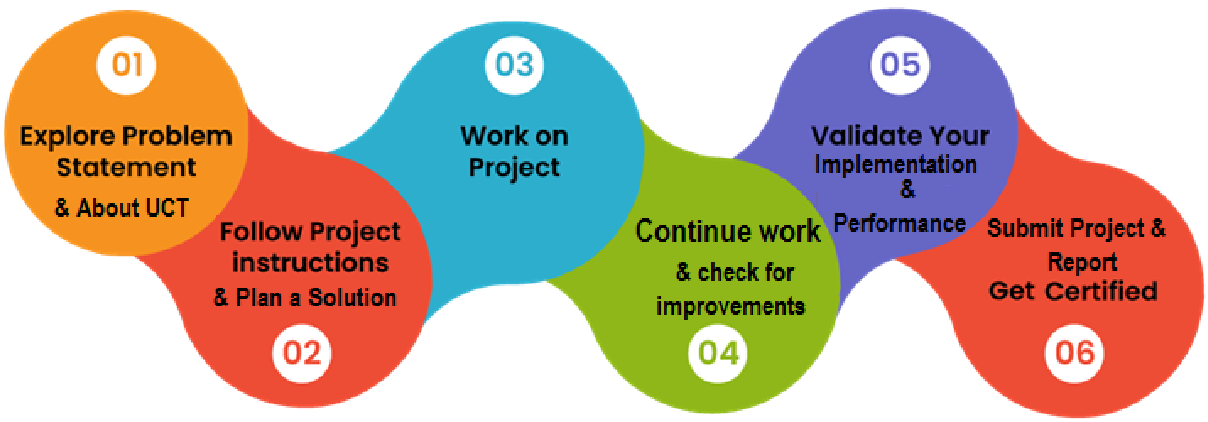
1 Preface

The internship was conducted over a period of six weeks, focusing on the application of Machine Learning techniques to solve real-world agricultural problems. The project assigned was “Prediction of Agriculture Crop Production in India.”

This project involved agricultural data analysis, data preprocessing, model building, and crop production prediction using Python and Machine Learning techniques. It provided practical exposure to predictive analytics and enhanced technical and problem-solving skills.

I would like to express my sincere gratitude to Upskill Campus, UniConverge Technologies Pvt. Ltd., and my mentors for their continuous support and guidance throughout the internship. This experience has significantly contributed to my technical and professional growth.



Your Learnings and overall experience.

Thank to all who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

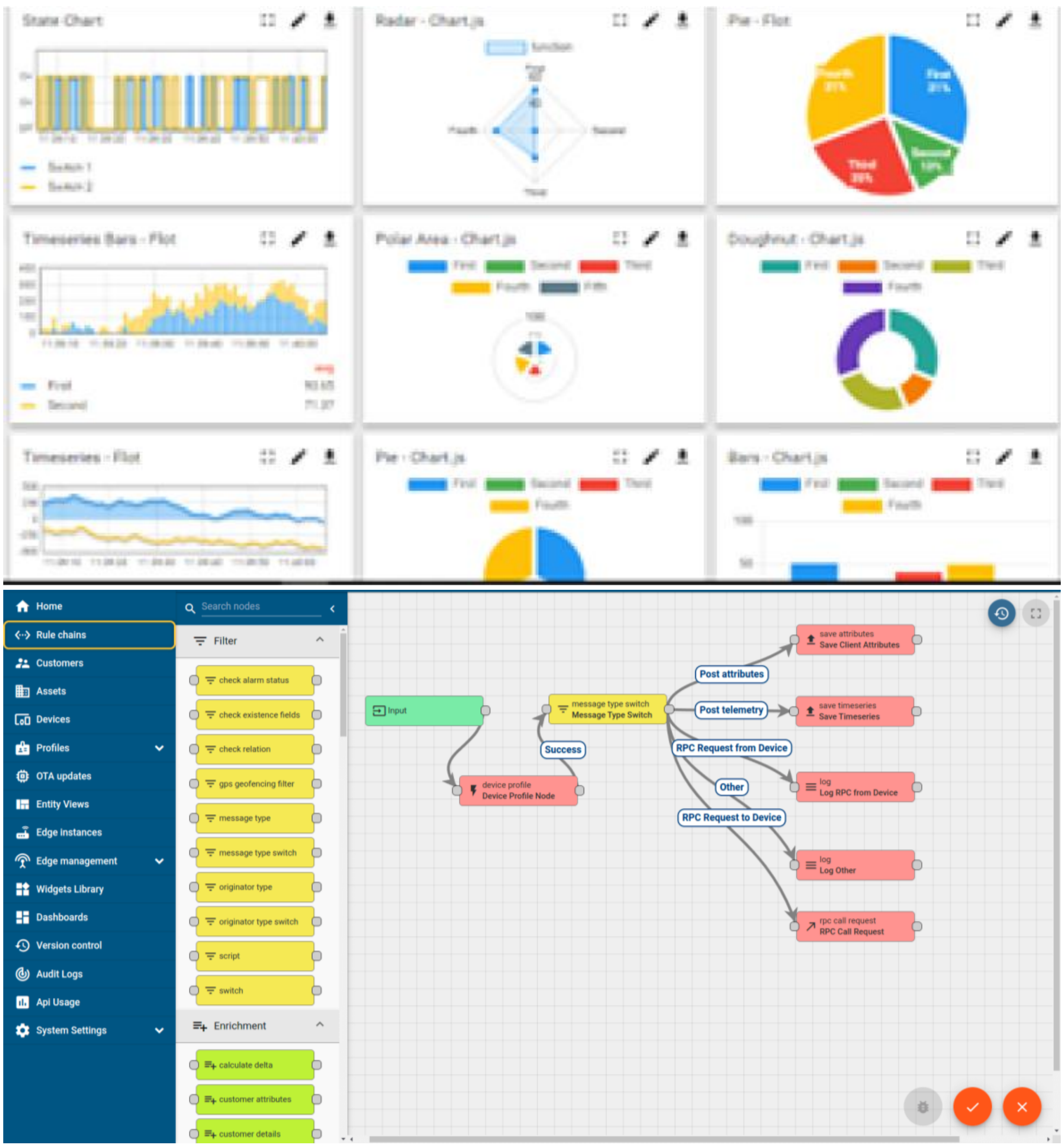
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

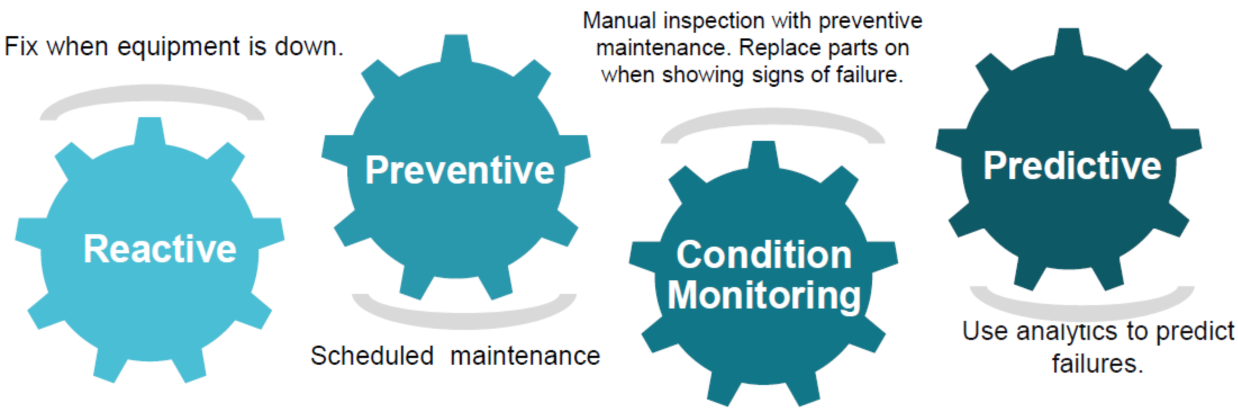
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

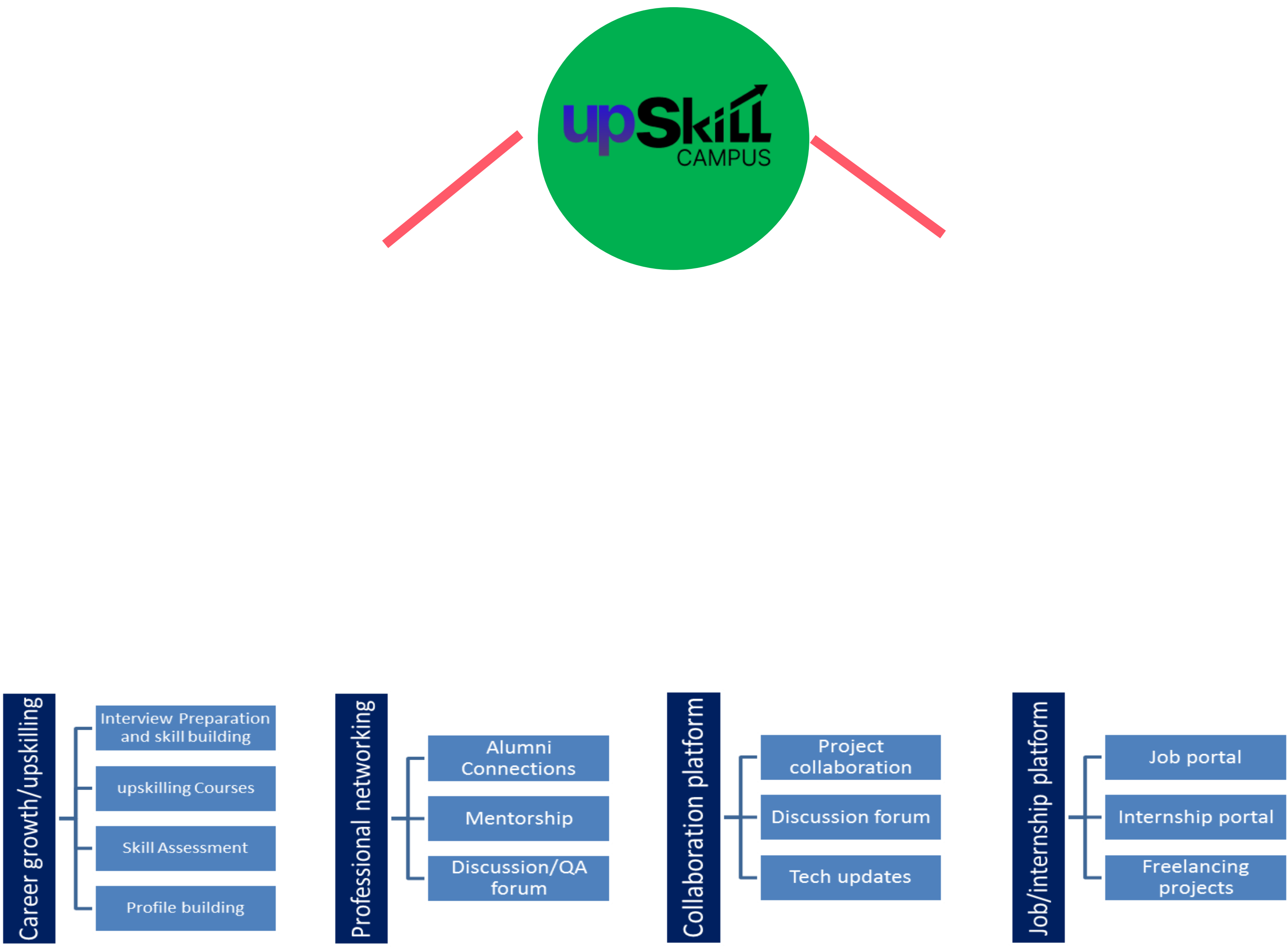
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ To gain practical experience in Data Science and Machine Learning.

➤ To analyze and process real-world datasets using Python.

➤ To develop predictive models using Machine Learning techniques.

➤ To improve problem-solving and analytical skills.

➤ To understand the practical applications of Data Science in real-world scenarios.

➤ To enhance technical knowledge and career opportunities in the field of Data Science and Machine Learning.

2.5 Reference

[1] Dataset provided by Internship Organization (Up Skills / UCT IoT Academy)

[2] Scikit-learn Documentation – https://scikit-learn.org

[3] Python Documentation – https://docs.python.org

[4] GitHub Repository -https://github.com/mohanabarkaviJB/upskillcampus

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Artificial Intelligence	AI
Random Forest	RF
Crop Production Prediction	CPP
Data Preprocessing	DP

3 Problem Statement

In the assigned problem statement

In this project, the main task is to develop a system for predicting agriculture crop production in India using machine learning techniques. However, the given dataset contains raw agricultural data which is not directly suitable for model building.

The problem faced during this phase is to properly handle and preprocess the dataset by cleaning missing values, handling categorical data, removing inconsistencies, and selecting relevant features.Without proper preprocessing, the machine learning model cannot give accurate predictions.

Therefore, it is necessary to transform the raw data into a structured format and prepare it for training the model to achieve better prediction results.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing systems for agricultural production prediction are mostly based on traditional methods and manual analysis. Farmers rely on past experience and historical data to estimate crop production. These methods often lack accuracy and do not effectively utilize Machine Learning techniques.

What is your proposed solution?

The proposed system uses machine learning techniques to predict crop production in India based on historical agricultural data. The dataset is preprocessed by handling missing values, removing inconsistencies, and selecting relevant features.

A machine learning model is then trained using the processed data to make accurate predictions. This system improves prediction accuracy and helps in better decision-making.

What value addition are you planning?

This project adds value by providing a data-driven approach for predicting agriculture crop production in India. It improves prediction accuracy, reduces manual effort, efficiently handles large datasets, and supports better agricultural planning and decision-making.

4.1 Code submission (Github)

https://github.com/mohanabarkaviJB/upskillcampus/blob/main/AgricultureCropPrediction.ipynb?short_path=a53fa6a

4.2 Report submission (Github link) : first make placeholder, copy the link.

https://github.com/mohanabarkaviJB/upskillcampus/blob/main/AgricultureCropProductionPrediction_MohanaBarkavi%20JB_USC_UCT.pdf

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

Agricultural Dataset

↓

Data Preprocessing

↓

Feature Selection

↓

ML Model

↓

Crop Production Prediction

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

Dataset

↓

Data Cleaning

↓

Handling Missing Values

↓

Feature Encoding

↓

Train-Test Split

↓

Model Training

↓

Prediction

↓

Performance Evaluation

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

User

↓

Input Form (Agricultural Data)

↓

System Processing

↓

Display Result (Crop Production Prediction)

Performance Test

The system performance depends on constraints such as memory, speed, and accuracy. The model uses moderate memory and provides fast prediction results. Data preprocessing techniques and efficient machine learning algorithms were used to handle these constraints effectively.

The test results show that the model achieves good accuracy and performs efficiently. The system can be further improved by using larger datasets and advanced algorithms to increase prediction accuracy and performance.

5.4 Test Plan/ Test Cases

TC01: Dataset Loading – Verify that the agricultural dataset is loaded successfully.

TC02: Data Preprocessing – Verify that missing values and inconsistencies are handled correctly.

TC03: Model Training – Verify that the machine learning model is trained successfully using the processed dataset.

TC04: Prediction Generation – Verify that the model generates accurate crop production predictions.

Result:

All test cases were executed successfully, and the system produced reliable crop production prediction results.

5.5 Test Procedure

The agricultural dataset was loaded into the system.

Data preprocessing was performed to handle missing values and inconsistencies.

The dataset was divided into training and testing sets.

The Machine Learning model was trained using the training data.

Predictions were generated using the testing data.

The model performance was evaluated based on prediction accuracy.

Result

The model successfully predicted agricultural crop production and achieved satisfactory performance.

5.6 Performance Outcome

The Machine Learning model successfully analyzed the agricultural dataset and generated accurate crop production predictions. The system performed efficiently after data preprocessing and model training. The results demonstrated good prediction accuracy and reliability, making the model suitable for agricultural data analysis and decision-making.

5.7 My learnings

During this internship, I gained practical knowledge in Data Science and Machine Learning. I learned how to collect, preprocess, and analyze real-world datasets using Python. I also gained experience in building and evaluating Machine Learning models for agricultural crop production prediction. This project improved my technical, analytical, and problem-solving skills and enhanced my understanding of predictive analytics and data-driven decision-making.

6.Future work scope

The future scope of this project is to improve prediction accuracy by using larger agricultural datasets and advanced Machine Learning algorithms. Additional parameters such as weather conditions, rainfall, soil characteristics, and seasonal factors can be included to enhance the prediction process. The system can also be developed into a user-friendly web or mobile application to help farmers and agricultural organizations make better decisions based on crop production forecasts.